

OBSERVE

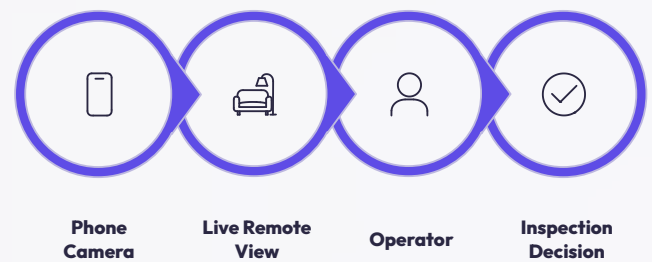
A REMOTE EYE INTO THE ENVIRONMENT.

The CyberRover X4 is equipped with a phone camera that provides live visual observation of the rover's surroundings. The operator views the camera feed in real time, enabling remote inspection of environments that are difficult or dangerous to access directly.

Current Visual Capabilities

- **Live Remote Camera Feed**
- **Visual Confirmation of Obstacles and Terrain**
- **Observation of Hazardous Locations**
- **Support for Navigation and Inspection Decisions**

Vision Pipeline



Future Development: A dedicated analog drone camera can replace or supplement the phone-camera arrangement in a future version, providing a more integrated onboard vision system with improved stability and field-of-view control.

THE OPERATOR

CONTROL WITHOUT DIRECT EXPOSURE.

The CyberRover X4 is controlled via a custom remote controller based on the ESP32 microcontroller. The operator remains in the decision loop, using ESP-NOW wireless control to direct the rover while receiving real-time telemetry and visual feedback. Commands flow from the controller through ESP-NOW to an Arduino Uno on the rover, which translates inputs into movement and sensor activation. This architecture keeps the human operator safe while extending physical access into hazardous environments.



ESP-NOW Wireless Link

2.4 GHz wireless protocol between controller and rover for reliable command transmission.



Joystick Input

Manual driving with precise directional control via physical joystick interface.



OLED Cyber OS Display

Onboard controller display showing system status and Cyber OS interface.



Real-Time Telemetry

Live sensor readouts delivered to the operator for informed environmental assessment.



Safety-Oriented Behavior

Control logic designed to prioritise safe operation in uncertain environments.



The operator maintains full situational awareness through the controller display and camera feed, making informed decisions about rover movement, sensor activation, and environmental assessment in real time.

DEPLOYMENT

WHERE CYBERROVER X4 CAN GO FIRST.



Industrial Inspection

Gas-leak detection, hazardous atmospheres, machinery areas, restricted zones.



Confined Spaces

Pipes, ducts, crawl spaces, tunnels, narrow passages.



Mining & Hazardous Environments

Remote inspection where gases, heat, or difficult terrain create additional risk.



Fire-Affected Areas

Remote visual and environmental assessment in or near fire-affected locations.



Earthquake & Disaster Response

Inspect damaged structures and debris zones before human responders enter.



Remote Reconnaissance

Observation of distant, uncertain, or difficult-to-access locations.



Note: Army and defence surveillance applications are potential use cases. The CyberRover X4 is not currently deployed in military operations or certified for battlefield use.

LIMITATIONS

EVERY ENGINEERING SYSTEM HAS BOUNDARIES.

This section details the inherent limitations of the CyberRover X4 in its current prototype stage. Understanding these boundaries is critical for safe and effective deployment, and for guiding future development towards a fully deployable system.

- | | |
|---|---|
| 1 | Environmental Sensing
MQ sensors provide prototype-level environmental indication. Gas readings must not be interpreted as certified gas concentration measurements unless calibrated and validated. Extreme environments may require industrial-grade sensors. |
| 2 | Temperature
DHT22 is suitable only within its specified operating range. It should NOT be presented as a sensor for 70–90°C environments. For extreme-temperature mining, industrial or fire environments, a future version requires an appropriately rated industrial temperature sensor. |
| 3 | Ultrasonic Sensing
Three ultrasonic sensors provide three-directional obstacle-distance information. They are read sequentially. Limitations include: measurement range, surface/angle dependence, environmental interference, blind spots, and inability to create a complete 3D map. |
| 4 | Visual Inspection
The current system uses a phone camera. Limitations include: dependence on the phone, field of view, lighting conditions, smoke/dust/obstruction, and mounting constraints. Future improvement: dedicated analog drone camera. |
| 5 | Communication
The rover depends on wireless communication. Potential limitations: range, interference, obstacles, signal degradation, loss of communication. The system should retain safe stop / operator safety behavior where implemented. |
| 6 | Mobility & Power
The rover's mobility is limited by: terrain, traction, obstacle size, wheel/chassis geometry, motor capability, battery capacity, and operating time. Do not invent numerical limits. |
| 7 | Autonomy & Intelligence
The current CyberRover X4 is operator-controlled. The ultrasonic system provides sensor-assisted navigation. AI, voice control, computer vision and higher-level autonomous decision-making are NOT implemented in the current version. These remain future development areas. |
| 8 | Safety & Certification
CyberRover X4 is a prototype engineering platform. It is NOT currently certified as: explosion-proof equipment, intrinsically safe equipment, mining-certified equipment, fire-entry equipment, or military equipment. Real deployment into regulated or hazardous environments would require appropriate certified hardware, testing and safety procedures. |

ENGINEERING STATUS

ENGINEERED IN LAYERS. BUILT TO EVOLVE.

Built & Demonstrated

- 4WD mobility
- 4 × geared DC motors
- BTS7960 motor drivers
- Remote controller
- ESP32 Dev Module
- ESP-NOW control link
- ESP32-S3 N8R16
- Arduino Uno motor/ultrasonic controller
- Arduino Nano environmental-sensor controller
- MQ-4, MQ-7, MQ-135
- DHT22 temperature/humidity
- 3× ultrasonic sensors
- Sequential ultrasonic measurement
- UART telemetry
- Wi-Fi dashboard
- Phone-camera visual
- OLED Cyber OS

Integration & Refinement

- System reliability
- Sensor integration
- Telemetry robustness
- Field testing
- Mechanical refinement
- Environmental testing
- Power management
- Wireless range
- Sensor calibration
- Durability validation

Future Development

- Industrial high-temp sensing
- Dedicated analog camera
- Integrated vision system
- Computer vision
- Voice interface
- AI command interpretation
- Autonomous navigation
- Advanced obstacle avoidance
- Extended sensor suite
- Modular payloads

1**Current Prototype****2****Field Validation****3****Hardened Hardware****4****Advanced Vision****5****AI / Autonomy**

MOVE. SENSE. INSPECT.

CYBERROVER X4

EXTENDING HUMAN AWARENESS INTO PLACES THAT ARE DIFFICULT, DANGEROUS OR
IMPOSSIBLE TO INSPECT DIRECTLY.
